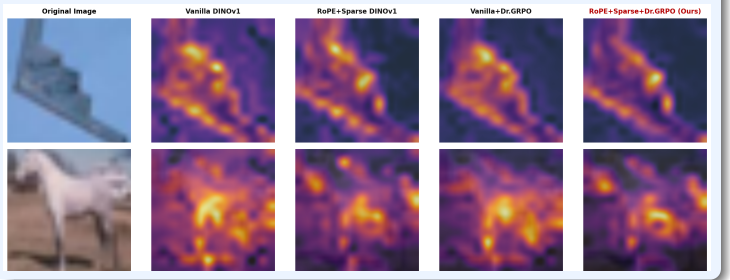
Verification Is All You Need

Emergent Object Discovery in Self-Supervised Vision Transformers via Dr. GRPO

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Master of Science in Data Science · **VNUHCM – University of Science**

Visual Teaser: Emergent Object Discovery via Dr. GRPO



Stage 1 (RoPE+Sparse Base): Continuous geometry resolves sharp local sub-parts.

Stage 2 (+ Dr. GRPO Alignment): RLVR purges background noise, boosting IoU by **+20.10%**!

+20.10%
Foreground Mask IoU

99.91%
Unsupervised CorLoc

0 Overhead
Exact 21.67M Params

< 3 GB
RLVR VRAM Footprint

Executive Highlights: Solving Vision Transformer Bottlenecks

The Fundamental Research Question:

- *Can Vision Transformers learn exact 2D geometry and discover object contours without human labels, without neural reward models, and with zero added parameters?*

The Dr. DINO Architecture Paradigm:

1. **Frequency-Adaptive 2D RoPE**: continuous relative geometry.
2. **Spatial-Semantic Sparse Attention**: -86.8% computation.
3. **Verifiable Dr. GRPO Alignment**: label-free RLVR.

+20.10%
Foreground Mask IoU

99.91%
Unsupervised CorLoc

0 Overhead
Exact 21.67M Params

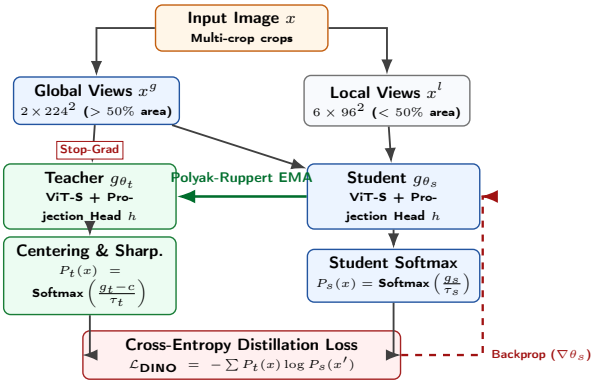
< 3 GB
RLVR VRAM Footprint

Core Breakthrough

83.98% Linear Probe Top-1 (+2.52%) and **75.97%** k-NN Top-1 (+1.56%) on CIFAR-10 with **exact parameter parity**!

The Foundation: DINOv1 Self-Distillation Framework

DINO (**DI**stillation with **NO** labels – Caron et al., Meta AI, ICCV 2021) revolutionized self-supervised learning:



Student–Teacher Dynamics

- **Identical Architecture**: Both networks share a ViT backbone and 3-layer MLP ($K = 65,536$ dims).
- **Online Momentum Teacher**: Updated strictly via Polyak-Ruppert EMA ($\lambda \in [0.996, 1.0]$) with **no backpropagation**.

Local-to-Global Induction

Teacher sees only **global scenes**; Student must predict the whole scene from **small local glimpses**.

DINOv1 Mechanics: Centering, Sharpening & Polyak EMA

Self-distillation aligns cross-view distributions **without negative pairs**. Stability relies on coupled dynamics:

1. Centering & Sharpening

Centering ($m = 0.9$, prevents mode collapse):

$$g_t(x) \leftarrow g_t(x) - \mathbf{c}, \quad \mathbf{c} \leftarrow m\mathbf{c} + (1-m)\frac{1}{B}\sum_{i=1}^{B} g_{\theta_t}(x_i)$$

Subtracting EMA center pulls dominant dims → uniform distribution.

Sharpening ($\tau_t = 0.04 \ll \tau_s = 0.10$, prevents entropy collapse):

$$P_t(x)^{(k)} = \frac{\exp\left((g_{\theta_t}(x)^{(k)} - c^{(k)})/\tau_t\right)}{\sum_{j=1}^{K}\exp\left((g_{\theta_t}(x)^{(j)} - c^{(j)})/\tau_t\right)}$$

Low temperature concentrates peaks → prevents uniform blur.

2. Dynamic Cosine EMA & Multi-Crop Loss

Polyak-Ruppert Cosine EMA ($\lambda_0 = 0.996 \rightarrow 1.000$):

$$\theta_t \leftarrow \lambda_t\theta_t + (1-\lambda_t)\theta_s, \quad \lambda_t = 1 - (1-\lambda_0)\frac{\cos(\pi t/T)+1}{2}$$

Smoothly slows teacher evolution to provide stable targets.

Multi-Crop Cross-Entropy (2 global x^g + 6 local x^l):

$$\mathcal{L}_{\text{DINO}} = \sum_{x\in\{x_1^g, x_2^g\}}\sum_{x'\in V, x'\neq x} -P_t(x)^\top \log P_s(x')$$

Student $P_s(x') = \mathrm{Softmax}(g_{\theta_s}(x')/\tau_s)$ updated via $\nabla\theta_s$.

The Breakthrough of DINOv1: Emergent Properties in ViTs

Prior to DINO, self-supervised learning flourished on CNNs but failed to scale cleanly on Vision Transformers:

1. Emergent Object Segmentation

- **Explicit Scene De-occlusion**: Self-attention maps of the last-layer [CLS] token segment physical object boundaries **completely without supervision**.
- **Head Specialization**: Different heads naturally capture foreground objects, textures, and background contours without any human bounding boxes.

2. High-Quality Semantic Clustering

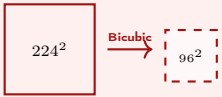
- **High-Performance k-NN**: Direct k-nearest neighbor classification on frozen ViT features achieves **78.3%** Top-1 on ImageNet-1K without fine-tuning.
- **Geometric Semantics**: Feature manifolds group semantically related concepts together.

The Unsolved Dilemma: Why DINOv1 Needs Enhancement

Despite this brilliance, standard DINOv1 ViTs suffer from **coordinate warping under multi-crop**, **quadratic attention dilution**, and **residual background halos**.

The Problem: 3 Structural Flaws in Vanilla DINOv1

1. Multi-Crop Warping



- Static APEs warped by interpolation.
- Destroys relative spatial phase.

2. Attention Dilution



- $> 85\%$ weights near zero.
- Background (sky/grass) pollutes `[CLS]`.

3. Background Leakage

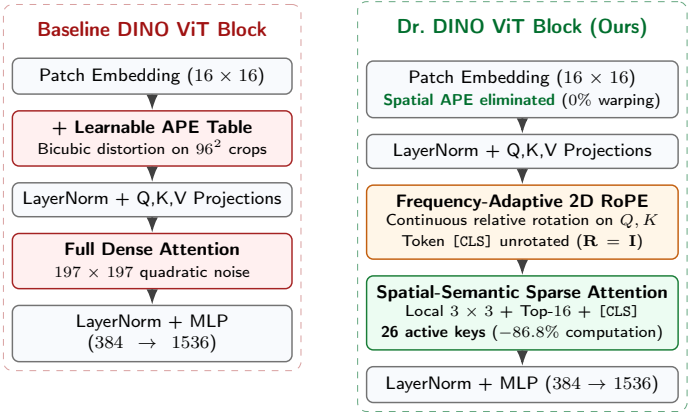


- Global CE ignores spatial masks.
- Residual noise persists indefinitely.

Dr. DINO Paradigm Shift

Continuous Geometry (2D RoPE) $\oplus$ **Focused Routing (S^3 Attention)** $\oplus$ **Verifiable RLVR (Dr. GRPO)**

Architectural Blueprint: Baseline vs. Dr. DINO Block



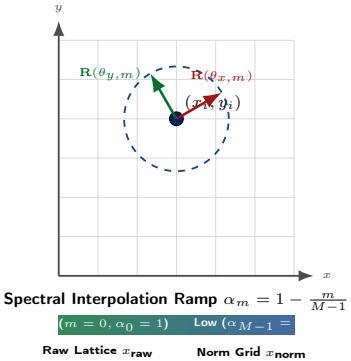
Strict Zero-Overhead Guarantee

- **Parameters**: Exactly **21,665,664** (0% parameter inflation).
- **FLOPs**: Exactly **9.20 GFLOPs** (identical theoretical complexity).
- **Translation Invariance**: Continuous spatial displacement $(\Delta x, \Delta y)$ directly encoded!

Scientific Rigor

Performance gains arise purely from geometric inductive bias, not model capacity inflation!

Innovation 1: Frequency-Adaptive 2D RoPE (Math Made Visual)



Spectral Coordinate Blending ($M = 16$ bands/axis)

$$\omega_m = B^{-2m/d_x}, \quad x_m^{\text{adapt}} = \alpha_m x_{\text{raw}} + (1-\alpha_m) x_{\text{norm}}, \quad \alpha_m = 1 - \frac{m}{M-1}$$

- **High Freq ($\alpha_0 = 1$)**: Preserves fine-grained local patch steps.
- **Low Freq ($\alpha_{M-1} = 0$)**: Scale-invariant global envelope across crops.

Orthogonal Rotation & Relative Positional Kernel

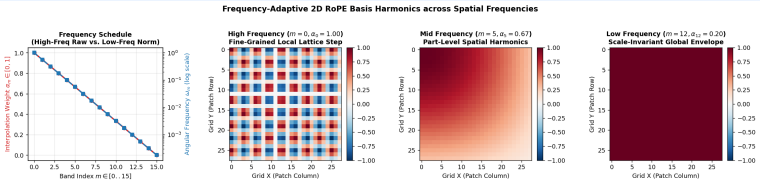
$$\mathbf{R}(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \quad \langle \mathbf{q}_i^{\text{rot}}, \mathbf{k}_j^{\text{rot}} \rangle = \mathbf{q}_i^\top \mathbf{R}(\Delta \mathbf{x}) \mathbf{k}_j$$

$$\mathcal{K}_{\text{RoPE}}(c,j) = 2 \sum_{m=0}^{M-1} \left[\cos\left(\Delta x_m^{\text{adapt}} \omega_m\right) + \cos\left(\Delta y_m^{\text{adapt}} \omega_m\right) \right]$$

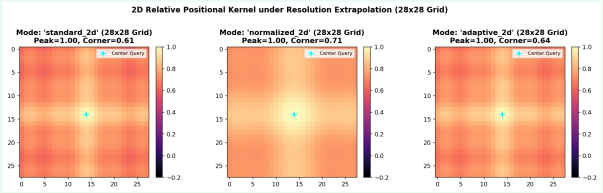
`[CLS]` **Token**: $\boldsymbol{\theta}_{\text{cls}} = \mathbf{0} \implies \mathbf{R} = \mathbf{I}$ (unbiased global aggregation).

Empirical Verification: 2D RoPE Harmonics & Extrapolation

1. Frequency Schedule & Spatial Basis Harmonics ($m = 0, 5, 12$)



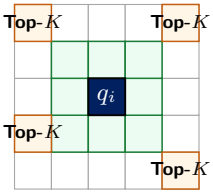
2. Relative Positional Decay Kernel under $14 \times 14 \rightarrow 28 \times 28$ Extrapolation



Innovation 2: Spatial-Semantic Sparse Attention (S^3)

Bidirectional [CLS] Conduit (Index 0)

$\mathcal{S}_0 = \{0 \dots N_p\}$ & $0 \in \mathcal{S}_i \ \forall i$



Composite Support Set $\mathcal{S}_i$ ($|\mathcal{S}_i| = 26/197$)

$$\mathcal{S}_i = \underbrace{\mathcal{N}_{\textbf{local}}(i)}_{3\times3\ \textbf{(9)}} \cup \underbrace{\text{Top-}16_{j \notin \mathcal{N}_{\textbf{local}}(i)}(\langle \mathbf{q}_i^{\textbf{rot}}, \mathbf{k}_j^{\textbf{rot}} \rangle / \sqrt{d_k})}_{\textbf{Dynamic Semantic (16)}} \cup \underbrace{\{\text{CLS}\}}_{\textbf{Global (1)}}$$

86.8% Key Reduction (13.2% active density).

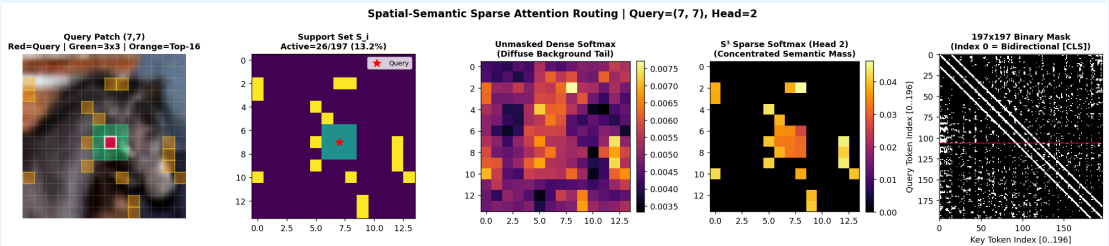
Exact Masked Softmax Normalization

$$A_{ij} = \frac{\exp(\tilde{s}_{ij})}{\sum_{l \in \mathcal{S}_i} \exp(\tilde{s}_{il})}, \quad \tilde{s}_{ij} = -\infty \ \forall j \notin \mathcal{S}_i, \quad \sum_{j \in \mathcal{S}_i} A_{ij} \equiv 1$$

Zero probability leakage $\implies$ $4\times$ higher peak foreground density!

Empirical Verification: S^3 Sparse Routing on Real Patches

Query Patch $(7, 7)$ on Horse (#24, Head 2): Local 3×3 + Dynamic Top-16 Keys



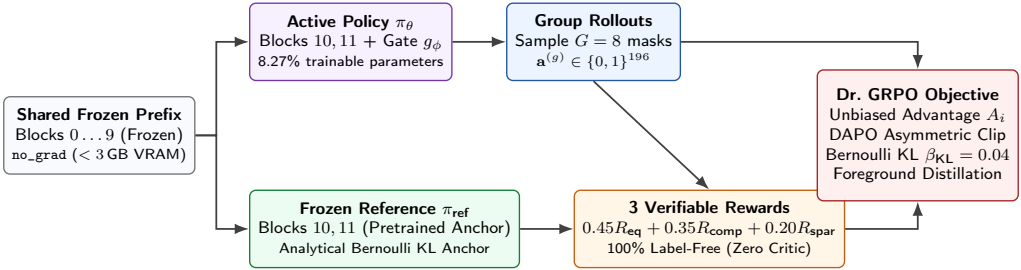
Unmasked Dense Softmax

Spreads probability mass across 197 keys (peak ≈ 0.0075), allowing background grass/sky clutter to dilute representations.

S^3 Sparse Softmax ($|\mathcal{S}_i| = 26$)

Eliminates 86.8% background keys and concentrates probability mass (peak > 0.040) strictly along the horse's anatomy!

Innovation 3: Verifiable RLVR via Unbiased Dr. GRPO



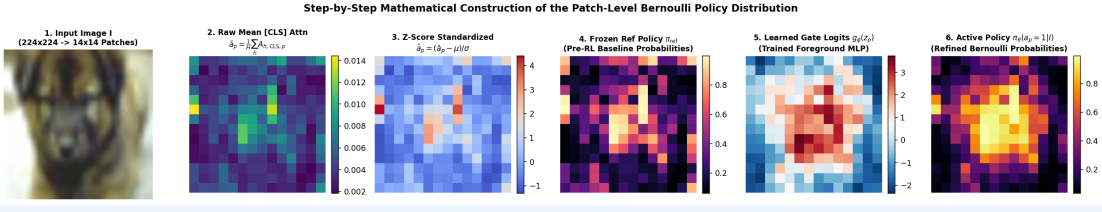
6-Step Bernoulli Patch Policy $\pi_\theta(\mathbf{a} \mid I) = \prod_{p=1}^{196} (p_p^\theta)^{a_p} (1 - p_p^\theta)^{1-a_p}$

$$\bar{a}_p = \frac{1}{H} \sum_{h=1}^{H} A_{h,\mathbf{CLS},p}^{(L)}, \quad \hat{a}_p = \frac{\bar{a}_p - \mu_{\bar{a}}}{\sigma_{\bar{a}} + \epsilon}, \quad b_{\text{sparsity}} = \log\left(\frac{0.35}{0.65}\right) \approx -0.619$$

$$\ell_p^\theta = 0.55\big(\gamma \hat{a}_p^\theta + b_{\text{sparsity}}\big) + 0.45\big(g_\phi(\mathbf{z}_p) + 1.35\,\hat{s}_p + b_{\text{sparsity}}\big), \quad p_p^\theta = \sigma(\ell_p^\theta)$$

Step-by-Step Construction of the Bernoulli Patch Policy

Empirical Policy Progression from Raw [CLS] Attention to Refined Active Policy π_θ



Frozen Pre-RL Policy $\pi_{\mathbf{ref}}$ (Step 4)

Exhibits strong background corner/border bias from static self-distillation, contaminating peripheral patches.

Active Policy $\pi_\theta(a_p = 1 \mid I)$ (Step 6)

Combining evolved Blocks 10–11 with learned residual gate $g_\phi(\mathbf{z}_p)$ suppresses border artifacts and locks onto the dog!

Verifiable Visual Rewards (`self_supervised_v1`)

1. Equivariance ($w = 0.45$)

$$\mathbf{m_{eq}} = \text{Top-}69\Big(\tfrac{1}{2}\left(T^{-1}(\pi_\theta(T(I))) + \mathbf{m_{sal}}\right)\Big)$$

$$R_{\mathbf{equiv}} = \frac{|\mathbf{a} \cap \mathbf{m_{eq}}|}{|\mathbf{a} \cup \mathbf{m_{eq}}| + \epsilon}$$

2. TokenCut + TV ($w = 0.35$)

$$R_{\mathbf{ncut}} = \frac{1}{2}\left(\frac{\mathbf{a}^\top \mathbf{W}(2\mathbf{a} - \mathbf{1})}{\mathbf{a}^\top \mathbf{W1} + \epsilon} + 1\right)$$

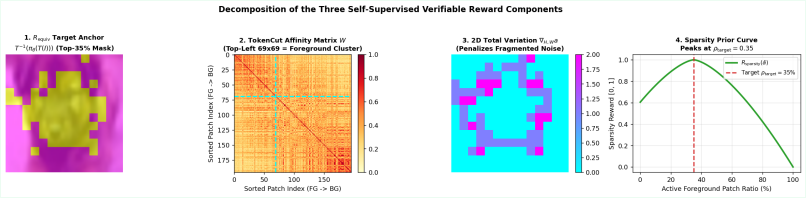
$$R_{\mathbf{comp}} = 0.65\, R_{\mathbf{ncut}} + 0.35\, R_{\mathbf{TV}}$$

3. Sparsity ($w = 0.20$)

$$R_{\mathbf{spar}} = \max\left(0,\ 1 - \left|\frac{\bar{a} - 0.35}{0.65}\right|^{1.5}\right)$$

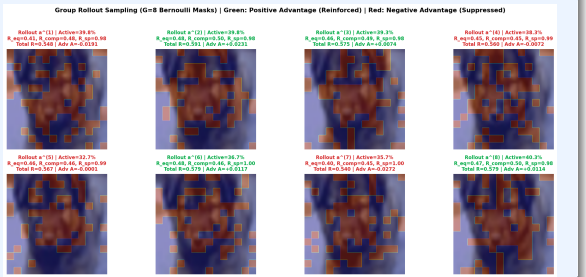
$$R = 0.45 R_{\mathbf{eq}} + 0.35 R_{\mathbf{comp}} + 0.20 R_{\mathbf{sp}}$$

Empirical Decomposition across All 4 Verifiable Signal Channels



Group Rollout Sampling ($G = 8$) & Unbiased Dr. GRPO Advantage

Sampled Group Rollouts ($\mathbf{a}^{(1)} \dots \mathbf{a}^{(8)}$)

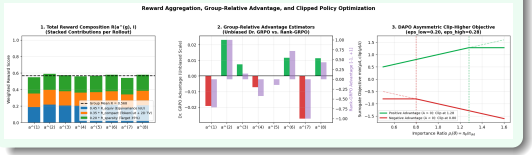


Unbiased Mean-Centered Advantage

$$A_i^{\textbf{Dr.GRPO}} = R^{(i)} - \frac{1}{G}\sum_{j=1}^{G} R^{(j)} \quad \text{(Zero } \sigma_R \text{ division bias!)}$$

$$\rho_i(\theta) = \exp\left(\frac{1}{N}\sum_{p=1}^{N} \log \frac{\pi_\theta(a_{i,p} \mid I)}{\pi_{\theta_{\textbf{old}}}(a_{i,p} \mid I)} + 0.1\,\Delta\mathcal{C}_{\textbf{align}}\right)$$

DAPO Asymmetric Clip-Higher



Stage-2 Complete Multi-Task Loss Formulation

1. DAPO Clipped Policy Loss $\mathcal{L}_{\text{clip}}(\theta)$

$$\mathcal{L}_{\text{clip}}(\theta) = -\frac{1}{G}\sum_{i=1}^{G} \min\Big(\rho_i A_i,\ \text{clip}(\rho_i, 1-\epsilon_{\text{low}}, 1+\epsilon_{\text{high}})A_i\Big)$$

$\epsilon_{\text{low}} = 0.20, \epsilon_{\text{high}} = 0.28$: asymmetric clip-higher encourages positive discovery while strictly suppressing background noise.

2. Analytical Bernoulli KL Divergence

$$\mathbb{D}_{\text{KL}}(\pi_\theta \parallel \pi_{\text{ref}}) = \frac{1}{N}\sum_{p=1}^{N} \text{kl}\left(p_p^{\theta} \parallel p_p^{\text{ref}}\right)$$

where $\text{kl}(p \parallel q) = p\log\frac{p}{q} + (1-p)\log\frac{1-p}{1-q}$.
Weight $\beta_{\text{KL}} = 0.04$: closed-form anchor with **zero variance**!

3. FG Distill & BCE Gate Alignment

Foreground Distillation ($w = 0.30$):

$$\mathcal{L}_{\text{FG-Distill}} = 1 - \left\langle \hat{\mathbf{f}}_{\text{cls}},\ 0.88\,\hat{\mathbf{f}}_{\text{cls}}^{\text{ref}} + 0.12\,\hat{\mathbf{f}}_{\text{fg}}^{\star} \right\rangle$$

Injects top spatial tokens $\hat{\mathbf{f}}_{\text{fg}}^{\star} = \text{Norm}(\sum_p a_p^{\star}\mathbf{z}_p)$ into `[CLS]`.
BCE Gate Alignment ($w = 0.25$, $\sigma_p^{\phi} = \sigma(g_\phi(\mathbf{z}_p))$):

$$\mathcal{L}_{\text{Gate}} = -\frac{1}{N}\sum_{p=1}^{N}\left[a_p^{\star}\log\sigma_p^{\phi} + (1-a_p^{\star})\log(1-\sigma_p^{\phi})\right]$$

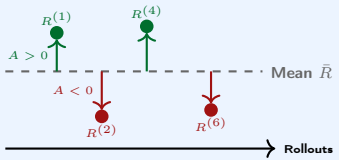
Total Stage-2 Composite Objective

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{clip}}(\theta) + \beta_{\text{KL}}\mathbb{D}_{\text{KL}} + 0.30\,\mathcal{L}_{\text{FG-Distill}} + 0.25\,\mathcal{L}_{\text{Gate}}$$

Dr. GRPO Unbiased Advantage & Convergence Dynamics

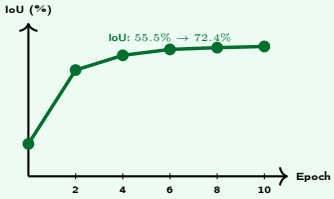
Unbiased Group Advantage

$A_i^{\text{Dr.GRPO}}$



- Exact zero-sum property: $\sum_{i=1}^{G} A_i^{\text{Dr.GRPO}} \equiv 0$.
- Zero σ_R division avoids exploding variance as $\sigma_R \to 0$.

10-Epoch Monotonic Convergence



- Rapid monotonic surge within first 2 epochs.
- Stable $\mathbb{D}_{\mathbf{KL}} < 0.08$ with zero catastrophic forgetting!

Quantitative Benchmark: CIFAR-10 Downstream Evaluation

Downstream evaluation of backbones pretrained for 190 epochs on **ImageNet-100**:

Metric	Baseline ViT-Small	Dr. DINO (Ours)	Gain (Δ)
k-NN Top-1 ($k = 10$)	74.41%	**75.97%**	**+1.56%**
k-NN Top-1 ($k = 20$)	74.34%	**75.31%**	**+0.97%**
k-NN Top-1 ($k = 200$)	70.41%	**70.65%**	**+0.24%**
Linear Probe Top-1 (Epoch 10)	80.92%	**83.38%**	**+2.46%**
Linear Probe Top-1 (Epoch 30)	80.85%	**83.63%**	**+2.78%**
Linear Probe Top-1 (**Best Checkpoint**)	81.46%	**83.98%**	**+2.52%**
Linear Probe Top-5 (**Best Checkpoint**)	99.31%	**99.44%**	**+0.13%**

Rapid Linear Convergence

By **Epoch 10**, Dr. DINO reaches **83.38%**, already beating the baseline's 100-epoch peak (**81.46%**)!

Exact Zero Capacity Inflation

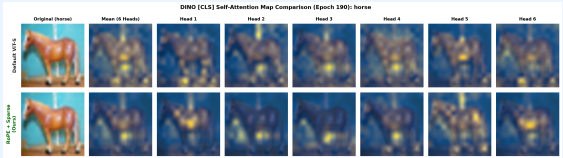
Both backbones have **21,665,664** parameters and **9.20 GFLOPs** ($\Delta = 0$). Gains stem purely from geometric bias.

Visual Evidence I: Stage-1 Multi-Head Attention Specialization

Airplane: Fuselage & Wings vs. Sky Clutter (Stage 1)



Horse: Muscle Contours vs. Turf Bleed (Stage 1)



Stage-1 Specialization

- **Baseline ViT**: Heads 1, 3, 4 diffuse mass onto sky and grass.
- **Dr. DINO-Base**: Heads 2 and 5 lock sharply onto physical structures.
- **No RLVR Yet**: Stage-1 backbones evaluated prior to Dr. GRPO post-training.

Complete 2×2 Factorial Ablation Matrix

Evaluated on full CIFAR-10 (50,000 train / 10,000 test images) across both axes:

Model Variant	Backbone Arch	Dr. GRPO	k-NN Top-1	Linear Probe	Mask IoU	CorLoc
1. Vanilla DINOv1	`vit_small`	No	71.13%	78.07%	58.88%	79.93%
2. Dr. DINO-Base	`vit_small_sparse_rope`	No	71.83%	80.22%	55.45%	69.26%
3. Vanilla + Dr. GRPO	`vit_small`	Yes	71.29%	78.44%	**78.98%**	**99.91%**
4. Dr. DINO-Full (Ours)	`vit_small_sparse_rope`	**Yes**	**72.05%**	**80.67%**	**72.44%**	**98.17%**

Massive Object Discovery Surge

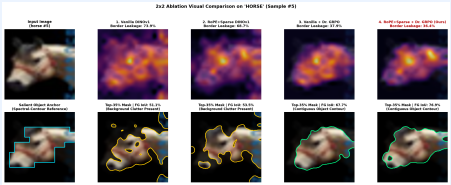
- **+20.10%** Mask IoU on Vanilla ($58.88\% \rightarrow 78.98\%$).
- **+16.99%** Mask IoU on Dr. DINO ($55.45\% \rightarrow 72.44\%$).
- **CorLoc reaches 99.91%** without any labels!

Synergistic Downstream Gains

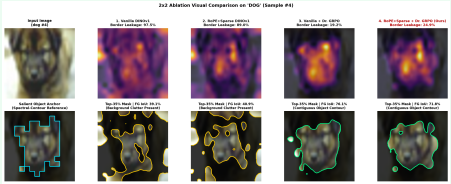
- **No Catastrophic Forgetting**: Downstream Linear Probe improves (+0.45% to **80.67%**).
- Top k-NN accuracy (**72.05%**) across all 4 variants!

Visual Evidence II: Contiguous Object Masks & Border Leakage

Horse (#5): Border Leakage $73.9\% \rightarrow \mathbf{36.4\%}$ | FG IoU $51.1\% \rightarrow \mathbf{76.9\%}$

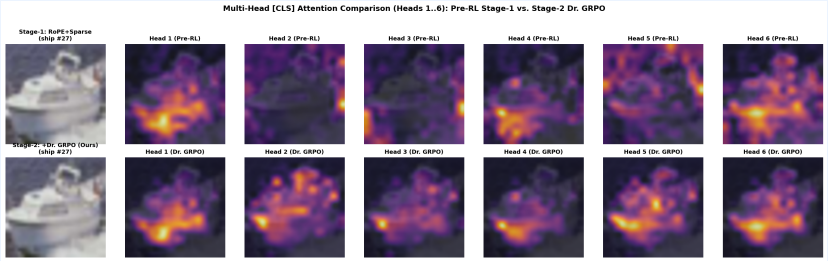


Dog (#4): Border Leakage $97.5\% \rightarrow \mathbf{19.2\%–24.9\%}$ | FG IoU $39.1\% \rightarrow \mathbf{76.1\%}$



Visual Evidence III: Pre-RL vs. Post-RL Multi-Head Alignment

Direct 6-Head [CLS] Comparison on ship (#27): Stage-1 vs. Stage-2 (+Dr. GRPO)



Stage-1 Pre-RL (RoPE+Sparse)

Heads 2, 3, and 5 suffer from perimeter border reflections and ocean background clutter (**Border Leakage = 97.9%, FG IoU = 32.9%**).

Stage-2 Post-RL (+Dr. GRPO)

All 6 heads purge perimeter noise (**Border Leakage = 33.4%**) and lock tightly onto the ship's hull and cabin (**FG IoU = 71.7%, +38.8% jump**)!

Hardware Profiling & Computational Efficiency

Benchmarked at 224×224 input resolution on an **NVIDIA RTX 3070 GPU**:

Resource Metric	Baseline ViT-Small	Dr. DINO (Ours)	Comparison
Backbone Trainable Parameters	21,665,664	21,665,664	**0% Overhead (Identical)**
Projection Head Parameters	9,720,064	9,720,064	Identical
Total Model Parameters	**31,385,728**	**31,385,728**	**31.39M (Identical)**
Theoretical FLOPs per Image	9.22 GFLOPs	9.22 GFLOPs	**Exact Parity (4.60 GMACs)**
Active Attention Pairs per Token	197 keys	**26 keys**	-86.8% **Sparsity Savings**
Inference Latency (BS=1)	1.20 ms	2.20 ms	+1.00 ms (PyTorch RoPE)
RLVR Policy VRAM Footprint	–	**< 3 GB**	**Shared-Prefix Efficiency**

Key Engineering Takeaway

Dr. DINO proves that advanced self-supervised vision models and RLVR post-training can be trained and evaluated efficiently on **consumer-grade GPUs** (<3 GB VRAM).

Conclusion: Three Pillars of Dr. DINO

1. 2D RoPE Geometry

- Continuous 2D rotation.
- Frequency-spectrum ramp.
- Eliminates multi-crop bicubic distortion.

2. Structured Sparsity

- Chebyshev 3×3 + Top-16 + [CLS].
- -86.8% key pairs.
- Exact Softmax normalization.

3. Verifiable Dr. GRPO

- Unbiased RLVR advantages.
- +20.10% Foreground Mask IoU.
- Reaches **99.91% CorLoc** without human labels!

Dr. DINO: Verifiable RLVR Discovery

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Thank you for your attention!

Questions & Discussion